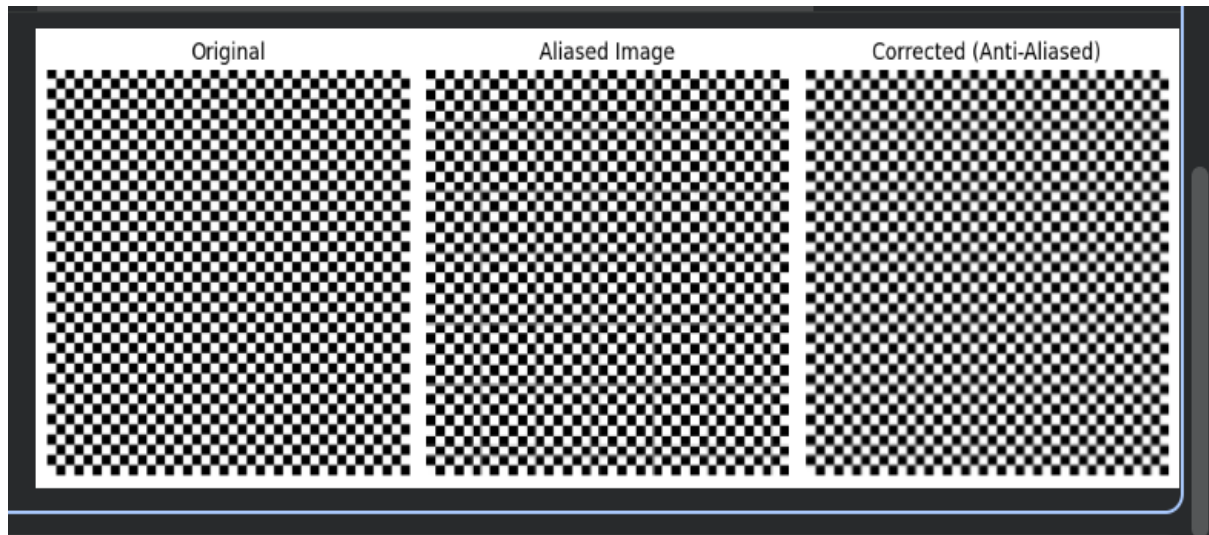


Aliasing Experiment - Step by Step Explanation

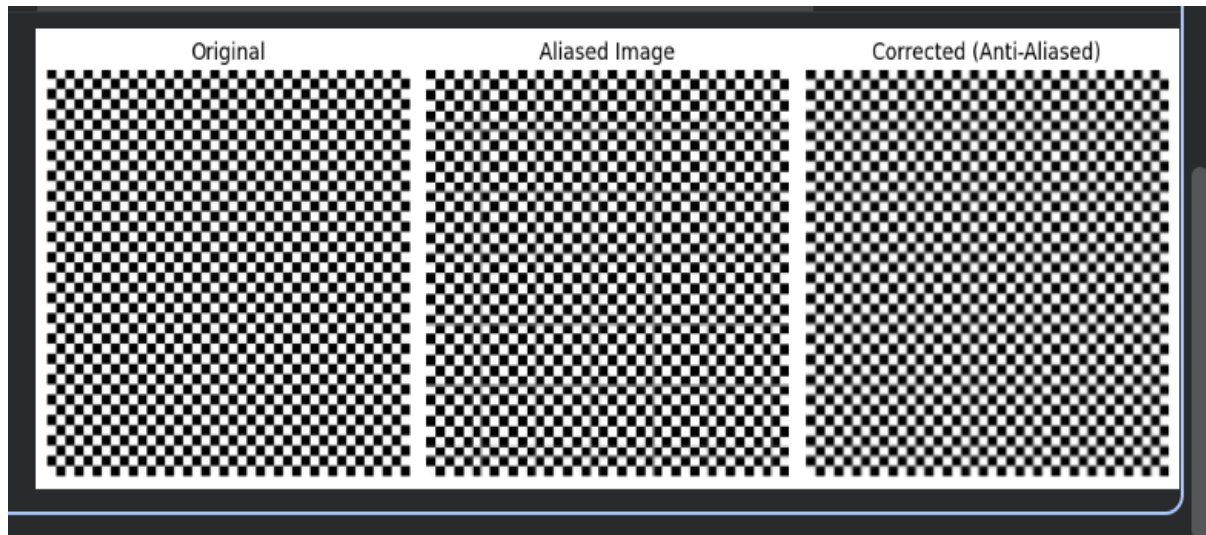
Step 1: Input Image

High-frequency checkerboard pattern is used as input. This contains fine details and sharp edges, which are ideal to demonstrate aliasing effects.



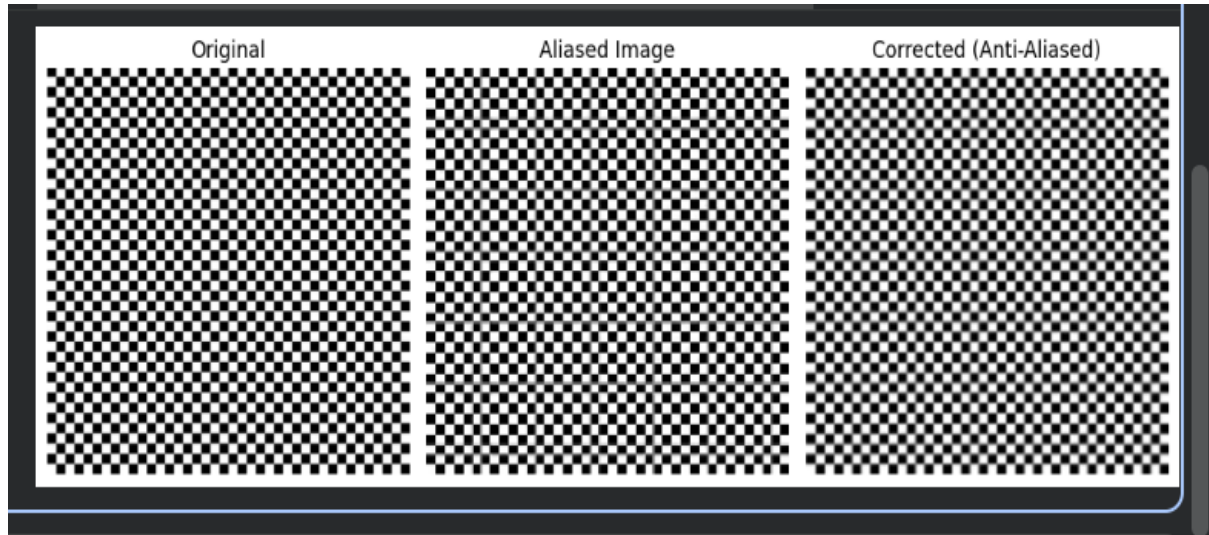
Step 2: Aliased Image (After Undersampling)

The image is downsampled without applying any filter. This violates the Nyquist condition, causing high-frequency components to overlap and produce distortion (aliasing).



Step 3: Corrected Image (Anti-Aliasing Applied)

A Gaussian low-pass filter is applied before downsampling. This removes high-frequency components and prevents aliasing. The image is then resized using bicubic interpolation for smooth output.



Conclusion

Aliasing occurs due to undersampling of high-frequency signals. It can be reduced by applying anti-aliasing filters before sampling and using proper interpolation techniques.